ONEACCESS

EKINOPS
Empowering Networks

DATASHEET

1424

SHDSL.bis Ethernet Access Device



The 1424 is OneAccess' Ethernet Access Device with SHDSL.bis EFM technology. It targets business customers with a need for high speed symmetrical bandwidth at various local loop distances. With its advanced layer 2 features and MEF certification, the 1424 meets all the requirements for access to Metro Ethernet networks. The device is interoperable with all major IP DSLAM vendors.

- SHDSL.bis EAD up to 4 pairs
- Connection speeds up to 22 Mbps over copper
- Advanced Layer 2 features
- EFM or ATM IMA pair bonding
- Extensive Maintenance and Management Functionalities
- MEF certified

High Speed DSL Access

The 1424 SHDSL.bis EAD targets business customers with a need for high speed symmetrical bandwidth in various DSL networks and at various local loop distances. It includes a SHDSL.bis interface up to 4 copper pairs. This provides line rates up to 22Mbps on short distances and up to 10Mbps on operator standard loop lengths. The various pair bonding techniques make it suitable for any type of DSL infrastructure. The symmetrical line rates of SHDSL perfectly match the traffic profiles of business users. Providing Ethernet in the First Mile over SHDSL.bis, the 1424 is interoperable with the major IP DSLAM vendors.

Extensive Feature Set

This access equipment has been designed to be used in a professional environment requiring advanced features such as native layer 2 support, advanced routing protocols (OSPF, BGP), advanced multilevel QoS support, several VPN implementations (IPSec, GRE, L2TP) and many management options. Backup scenarios are available including 3G backup via an external USB dongle. This eliminates the need to install additional router equipment to deliver Internet or IP VPN services to business customers.
With a bidirectional forwarding performance of 250Kpps, the 1424 SHDSL.bis EAD easily handles all layer 2 and 3 services at 22 Mbps.

Metro Ethernet Services

Ethernet features include many bridge cache options, STP, RSTP, MSTP, advanced VLAN features and Ethernet OAM.

Quality of Service is available both on IP and Ethernet packets with filtering, classification and colouring based on layer 2 and layer 3 characteristics, many priority scheduling options, policing and shaping. The 1424 fully complies with the MEF.9 and MEF.14 requirements as defined by the Metro Ethernet Forum.

Smooth Migration from ATM to EFM Networks

Operators gradually migrate from ATM based access networks to Ethernet based access networks. The 1424 supports both network types. This means that a 1424 can be installed in an existing ATM network. When the network migrates to Ethernet, the same 1424 can be re-used in the Ethernet network without on-site intervention. The unique feature of combining ATM and EFM access in the same device leverages a smooth migration of the access network.

SHDSL.bis Ethernet Access Device

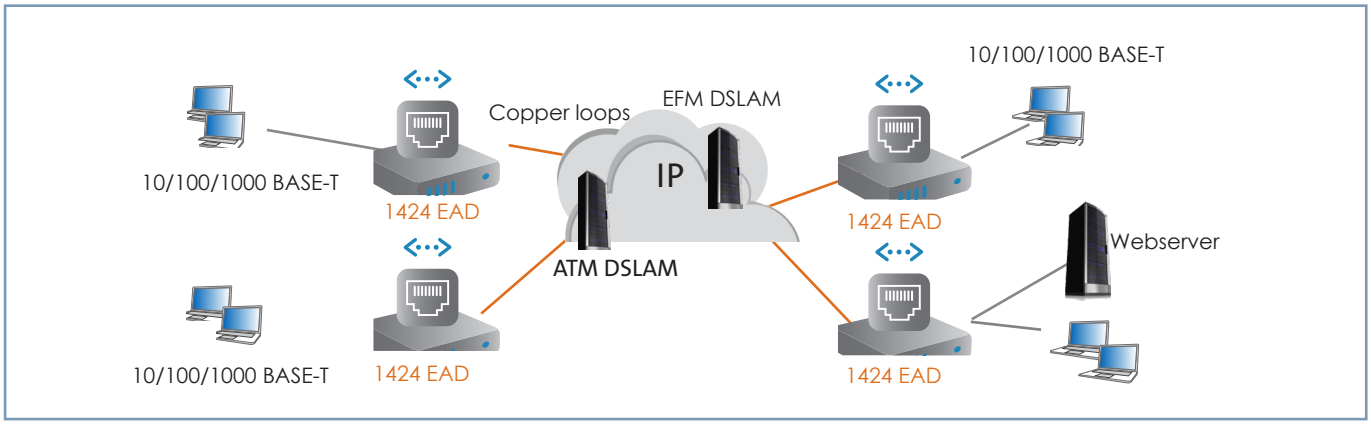
Accelerated Deployment and Service Provisioning

The OneAccess 1424 Fibre EAD can be integrated in any managed environment and supports all the common management interfaces such as SNMP, Telnet, SSH, HTTP and HTTPS. In addition to these interfaces a number of management tools are available to facilitate the integration of these access devices in a managed environment. These include:

- TMA GUI application
- A customisable Web-configuration utility
- A CLI for scripting and simple integration with provisioning and management systems
- Element and Inventory Management for control and monitoring of large access neworks

Quality Monitoring and Service Level Agreements

The 1424 keeps statistics of the last 2 hours, 24 hours and 7 days. Selected statistics can be stored over a longer period on the device for later retrieval and processing on a management platform. Traffic quality monitoring provides pro actively all the information to offer Service Level Agreements to the customer.



Technical Specifications





Basic Hardware

- G.SHDSL.bis line interface: 1/2/4 pairs
- 1 Fast Ethernet uplink
- 4 port Fast Ethernet VLAN switch
- 1 console port
- IPsec encryption accelerator for DES, 3DES and AES

SHDSL Interface

- G.SHDSL.bis1/2/4 line pairs
- RJ-45 connector
- ATM, EFM and ATM/EFM auto detect
- Bonding based on EFM, SHDSL with ATM and ATM IMA
- Synchronisation LEDs
- IEEE 802.3 2BASE-TL (aka 802.3ah) (for EFM operation)

Copper Ethernet Interfaces

- 4 port switch 10/100BASE-T
- Independent 10/100Base-T port
- Half/full duplex with auto-sense, automatic cross-over
- Link status and activity LEDs

USB Port (option)

- USB 2.0 compliant at USB 1.1 speed
- Support for mobile USB devices

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Technical Specifications



Layer-2 Functionality

- IEEE 802.1D Transparent Bridging
- IEEE 802.1D Spanning Tree Bridging
- IEEE 802.1W Rapid Spanning Tree Bridging
- IEEE 802.1S Multiple Spanning Tree Bridging
- IEEE 802.1Q VLAN Tagging
- IEEE 802.1p QOS on MAC level
- Basic and extended MAC filtering
- VLAN Switching
- Port-based VLANs
- IEEE 802.1ad Provider Bridges or QinQ
- OAM IEEE 802.3 chapter 57 (for EFM operation)
- OAM IEEE 802.1 ag & ITU-T Y.1731
- MEF9 and MEF14 certified

IP Functionality

- NAT/PAT
- DHCP client/relay/server
- DNS server/relay
- IGMPv1, v2
- Stateful Inspection Firewall
- Basic and Extended IP filtering
- DMZ

IP Routing

- Static routing
- Policy based routing
- RIP v1 and v2, OSPF, BGP-4
- VRF (VPN Routing & Forwarding)
- VRRP (Virtual Router Redundancy Protocol)

Virtual Private Networks

- GRE tunnelling
- L2TP tunnelling
- IPSec (tunnel and transport mode)
- GRE or L2TP transport mode
- IKE and Manual Key Management
- AH and ESP Protocol
- DES, 3DES and AES encryption
- SHA-1 and MD5 Authentication

Quality of Service (QoS)

- Traffic Classification and Policing (inbound/outbound)
- Priority Queuing Layer 3 (8 levels)
- Priority Queuing Layer 2 (8 levels)
- Traffic Shaping CIR/EIR
- Queuing mechanisms: SP, RR, WFQ, LDWFQ

Performance and Scalability

- Routing and bridging performance: 250 Kpps
- Number of IPSEC, L2TP or GRE tunnels: 25
- Number of VLANs: 200
- Number of bridge-groups: 13

Maintenance and Management Support

- Console port, CLI, Telnet, SSH
- Multilevel password protection, Radius AAA
- HTTP, HTTPS, customised Web Interface
- FTP/TFTP upload/download configuration/firmware
- SNMPv1, v2, v3, MIB II, proprietary MIB
- Statistics 15min, 2h, 24h, 7 days
- IP traffic monitoring: roundtrip delay, jitter, loss
- Syslog, SNTP
- PC-based maintenance tool
- Element management application
- Inventory management application

Dimensions

- Desktop, metal housing, wall mountable
- W x H x D: 275 x 65 x 145 mm (9.64 x 2.56 x 5.7 in)
- Weight: 0,8 kg (1.76 lb)

Power Supply

- External adapter 12V – 2A
- Voltage range: 110-230 VAC – 50/60 Hz
- Power consumption: <24 W

Ordering codes	Product description
208595	1424 SHDSL 1P NPWR
501426	1424 SHDSL 2P NPWR
208602	1424 SHDSL 4P NPWR
70538	1424 SHDSL 1P USB NPWR
70539	1424 SHDSL 2P USB NPWR
70540	1424 SHDSL 4P USB NPWR

About EKINOPS



Ekinops is a leading provider of open and fully interoperable Layer 1, 2 and 3 solutions to service providers around the world. Our programmable and highly scalable solutions enable the fast, flexible and cost-effective deployment of new services for both high-speed, high-capacity optical transport networks and virtualization-enabled managed enterprise services

Our product portfolio consists of three highly complementary product and service sets: Ekinops360, OneAccess and Compose.

- Ekinops360 provides optical transport solutions for metro, regional and long-distance networks with WDM for high-capacity point-to-point, ring and optical mesh architectures, and OTN for improved bandwidth utilization and efficient multi-service aggregation.
- OneAccess offers a wide choice of physical and virtualized deployment options for Layer 2 and Layer 3 access network functions.
- Compose supports service providers in making their networks software-defined with a variety of software management tools and services, including the scalable SD-WAN Xpress.

As service providers embrace SDN and NFV deployment models, Ekinops enables future-proofed deployment today, enabling operators to seamlessly migrate to an open, virtualized delivery model at a time of their choosing.

A global organization, with operations in 4 continents; Ekinops (EKI) - a public company traded on the Euronext Paris exchange - is headquartered in Lannion, France, and Ekinops Corp., a wholly-owned subsidiary, is incorporated in the USA.

EKINOPS360 Dynamic Optical Transport

COMPOSE

ONEACCESS Fast Network Virtualization

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